

DART-RIFT. I.

Cohesion, Matrix Tension, and Impact-Site Sensitivities in Simulations of Dimorphos

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Abstract. Five completed, validity-checked SPH pilot simulations were run in a fixed DART impact scene to measure model sensitivity before production inference. At matrix cohesion $Y_0 = 10, 500$, and 5000 Pa, simulated momentum enhancement is $\beta = 3.7480, 2.6247$, and 1.8738 , respectively; the corresponding recorded ejecta masses are $1.964 \times 10^7, 3.649 \times 10^6$, and 7.516×10^5 kg. At 10 Pa, enabling the tested matrix-tension law reduces β to 1.0807 , while a 25 m impact offset gives 3.6650 and changes model ejecta mass by 18% . A three-point power law for $\beta - 1$ has slope -0.1791 and maximum relative residual 0.1198 ; its scene-conditioned interpolation gives 20.1 Pa at the mass-adjusted observed β . That interpolation is not an interior-strength measurement or posterior. We register a 96-point coarse design and 12 nested fine reruns, and identify unresolved numerical and observation-definition checks that prevent a defensible production posterior or numerical Hera forecast at present.

1 Introduction

The DART experiment measured a change in Dimorphos’s orbit, constraining the momentum transferred to the body [1, 2]. Translating that response into interior structure requires a forward model whose material laws, target geometry, resolution, and observation operators are controlled. We use DART-RIFT to study three interior coordinates: block distension α_b , matrix cohesion Y_0 , and block volume fraction f . A match by one simulated scene is evidence about that scene, not an identified three-parameter interior.

This paper reports the completed pilot measurements first. It then specifies the production design and claim checks that were fixed before reading the production pool. The pilot runs themselves predate this paper and are not presented as blind predictions.

2 Simulation and comparison

2.1 Fixed scene and controlled changes

The reference target is a triaxial ellipsoid with semiaxes (88.5, 87, 58) m and calibrated mass 4.2980×10^9 kg (Figure 3). The three-sphere impactor has mass 579.4 kg, speed 6144.9 m s^{-1} , and nominal incidence 17° from the surface normal. The pilot cohesion comparison fixes $(\alpha_b, f) = (1.15, 0.25)$, the random seed 20260906, 7 m particle spacing, a 1.0 s transition time, and a 600 s endpoint [pilot protocol]. Figure 4

Table 1. Completed pilot runs at 600 s. Conditions other than the listed change are held fixed.

Run	Change	β	M_{ej} (kg)
DY2	$Y_0 = 10$ Pa	3.7480	1.964×10^7
DO1	$Y_0 = 500$ Pa	2.6247	3.649×10^6
DO2	$Y_0 = 5000$ Pa	1.8738	7.516×10^5
DC	Tension on	1.0807	2.777×10^2
DN	Offset 25 m	3.6650	2.321×10^7

places those two fixed times inside the sequence a run actually integrates. DO1 and DO2 change only Y_0 from the DY2 reference. DC enables one selected matrix-tension law at 10 Pa; DN moves the impact point 25 m at 10 Pa. The production model keeps that tension law disabled [model decision].

All four new branches completed and passed the repository’s validity checks; the reference DY2 also passed. Table 1 reports their recorded β and ejecta mass. The run identifiers, code commits, and file hashes are in the [run source record]. We use *model ejecta mass* for the simulation quantity; it is not automatically the same as an observed plume mass.

The four new branches have energy deviations between -0.79% and -0.80% , mass-consistency deviations between -0.05% and -0.06% , and momentum-ledger residuals no greater than 9×10^{-15} in the recorded checks [validation record]. These checks support run validity; they do not establish resolution convergence.

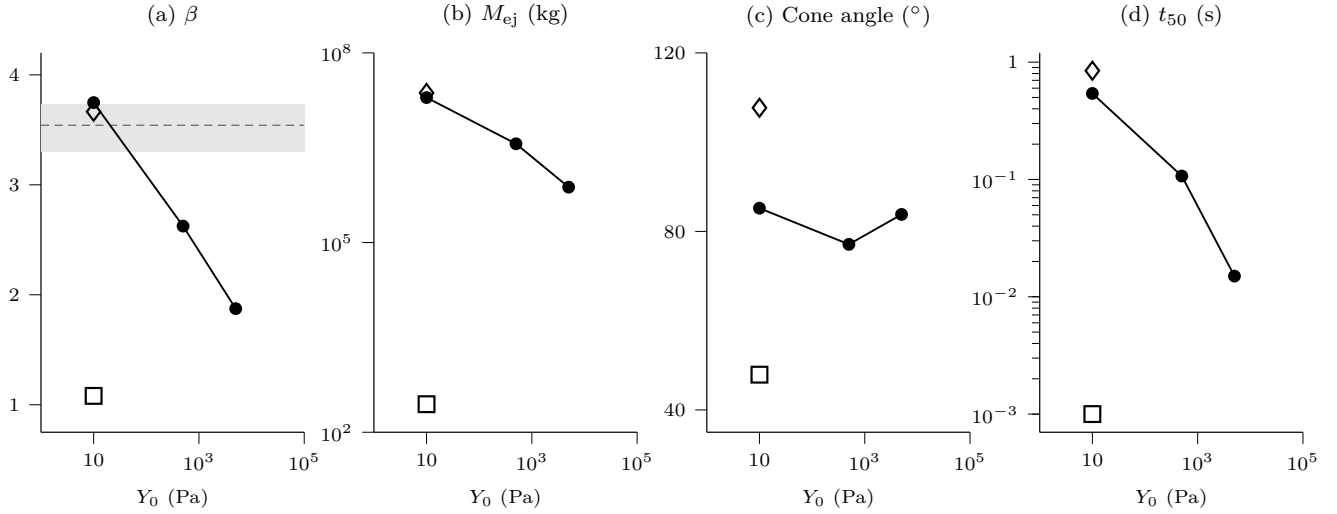


Figure 1. Four-observable atlas for the five completed pilot runs. Filled circles and connecting segments show only the three cohesion runs (DY2, DO1, DO2); the segments guide the eye and are not an inferred response curve. The open square is DC (matrix tension on) and the open diamond is DN (25 m impact offset); both retain $Y_0 = 10$ Pa but are not part of the cohesion sweep. Cone opening and median ejecta time t_{50} are model diagnostics. The gray band in panel (a) is the mass-adjusted DART 1σ interval from Equation 1; its dashed line is the central value. Panel (b) has no observational band because the model ejecta statistic is not the LICIAcube plume estimate.

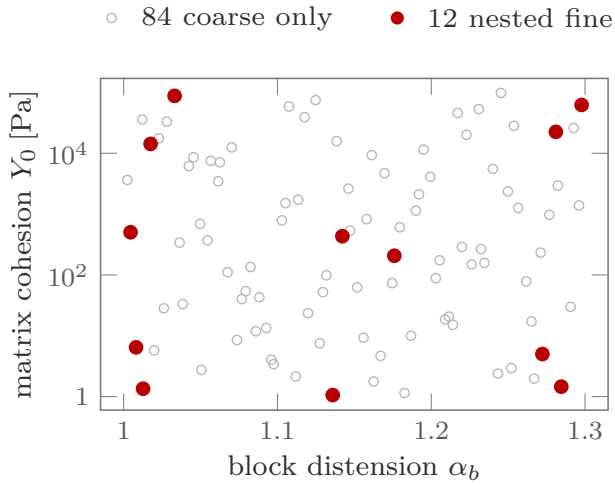


Figure 2. Locked design before production-pool inspection, shown in the (α_b, Y_0) face; f is sampled on the same hypercube over $[0.05, 0.50]$. Open circles are the 84 coarse-only points and filled circles the 12 nested fine locations, which were fixed by a deterministic rule and pinned by a test before any pool output existed. This is a design plot: it carries no measured response and no inferred interior.

2.2 Observational target

The mass-adjusted momentum target used for the production scene is

$$\beta_{\text{obs}}(M_{\text{scene}}) = 3.5418^{+0.188}_{-0.247} \quad (M_{\text{scene}} = 4.2980 \cdot 10^9 \text{ kg}). \quad (1)$$

The asymmetric interval is retained when constructing the likelihood [mass adjustment]. It is shown in Figure 1 only as

an observational reference; it is not a posterior interval.

3 Measured pilot results

3.1 Cohesion sweep

Across the tested 10–5000 Pa span, β falls from 3.7480 to 1.8738 while model ejecta mass falls by a factor of 26.1 (Figure 1). The predeclared three-point fit $\beta - 1 = CY_0^p$ gives $p = -0.1791$ and a maximum relative residual of 0.1198, below the fixed 0.15 fit-quality threshold [DO protocol]. At the central target in Equation 1, this fit interpolates to $Y_0 \approx 20.1$ Pa (Figure 5). The value is conditional on the fixed scene, α_b , f , material law, and fit form; it is not a posterior estimate of Dimorphos’s strength. The older production cohesion domain $[10^3, 10^7]$ Pa does not contain this conditional interpolation, motivating the prospectively selected $[1, 10^5]$ Pa domain.

3.2 Matrix tension

At $Y_0 = 10$ Pa, enabling the tested tension law changes β from 3.7480 to 1.0807. The contrast in excess momentum $b = \beta - 1$ is

$$\frac{|b_{\text{DY2}} - b_{\text{DC}}|}{b_{\text{DY2}}} = \frac{|2.7480 - 0.0807|}{2.7480} = 0.9706. \quad (2)$$

Recorded model ejecta mass changes from 1.964×10^7 kg to 2.777×10^2 kg, a factor of about 7.1×10^4 . This is a strong sensitivity to one constitutive-model choice. Because DC is one toggle at one parameter location, it does not establish the real body’s tensile strength or rule out all alternative tension laws. The production model choice remains explicit, and the DC

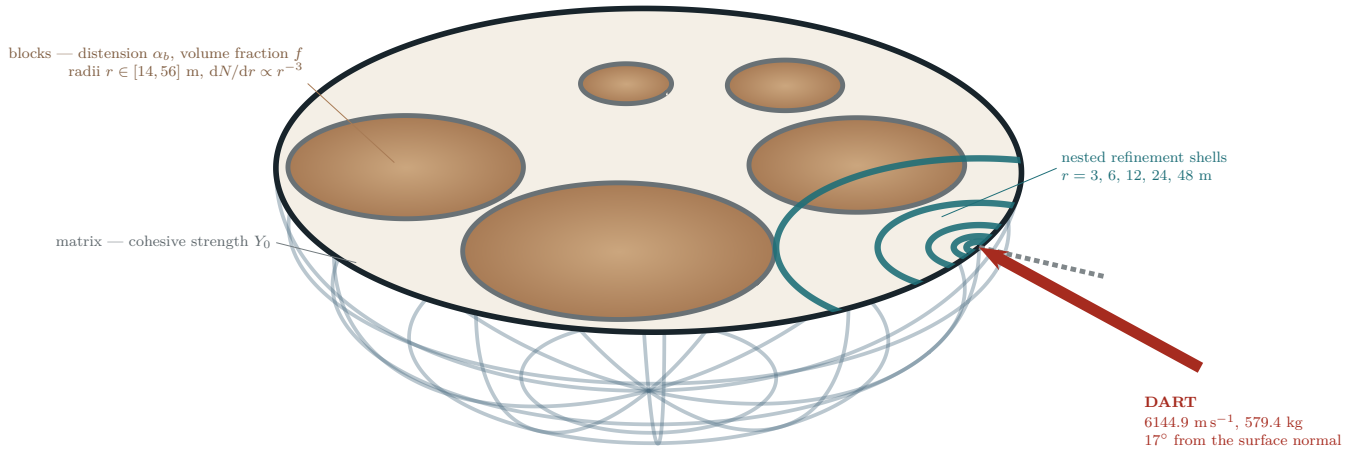


Figure 3. The fixed pilot and production scene, cut at the equatorial plane $z = 0$ and drawn from the scene builder’s own output rather than sketched. The lower half is a wire frame; the cut face carries the five interior blocks that the production block generator places at $f = 0.25$. The cut plane was chosen because all five blocks intersect it, whereas the meridional plane $y = 0$ intersects only one. Shell radii mark the nested spacing ladder about the impact point.

contrast is reported outside the small-discrepancy denominator [locked pilot judgments].

3.3 Impact-site offset

Moving the impact point 25 m changes β from 3.7480 to 3.6650 (2.2%), but changes model ejecta mass from 1.964×10^7 to 2.321×10^7 kg (18%). The recorded cone opening changes from 85.2° to 107.7° , and median ejecta time t_{50} from 0.541 to 0.845 s. A near-stable β therefore coexists with a materially different ejecta field in this controlled pair. The corresponding impact-location discrepancy on b is 0.03019; it exceeds the preregistered *negligible* boundary of 0.03000 by 0.00019. We retain the registered classification rather than moving that boundary after seeing the result [KAYIT-076]. The same pair is one row of the lever ranking in Figure 7.

4 Registered production inference

4.1 Design and validation

The production coordinates are $\theta = (\alpha_b, Y_0, f)$, with $\alpha_b \in [1.00, 1.30]$, $Y_0 \in [1, 10^5]$ Pa, and $f \in [0.05, 0.50]$. The design has 96 Latin-hypercube coarse points at 7 m spacing and 12 predetermined nested reruns at 3.5 m spacing [nested-design decision]. Figure 2 shows coverage; it is a design plot, not a fitted posterior. The 12 indices are {11, 20, 33, 44, 45, 57, 58, 62, 74, 78, 87, 92}. Every readable run must be valid, reach 600 s, and meet the 5% mass-consistency limit. A loss above 20% invalidates the pool and triggers its repetition [pool protocol].

The registered surrogate is a Gaussian process validated in four groups by θ , with predictive variance enlarged when held-out residuals require it. The residual-correlation matrix R is measured from those residuals and compared with $R = I$. The alternative resolution paths are a coarse surrogate plus

measured discrepancy (SIGMA), and an autoregressive nested coarse/fine surrogate (CD). CD leads only if the nested design is intact, the fitted coupling is interior, and held-out fine error is lower than for SIGMA. Both results are disclosed if they differ materially [posterior definition]. For eligible observation vector z and surrogate mean $g(\theta)$, the registered heteroskedastic likelihood is

$$\mathcal{L}(\theta) \propto |S(\theta)|^{-1/2} \exp\left[-\frac{1}{2}(g - z)^T S(\theta)^{-1}(g - z)\right]. \quad (3)$$

The covariance $S(\theta)$ combines predictive, observation, and admitted model uncertainty with the measured residual correlation. The determinant term is retained; no production posterior is evaluated in this report.

4.2 Observation mapping and model error

A LICIACube analysis reports an extrapolated plume mass $(1.6 \pm 0.3) \times 10^7$ kg as a plausible upper limit for mass within the camera field of view [3]. The model quantity in Table 1 is the reported ejecta statistic at 600 s; the run record also tracks a separate escaping-mass diagnostic. Neither is definitionally identical to the observed plume mass. Neither resolves a second threshold either: the body’s own escape speed follows from the scene’s mass and effective radius, 0.0865 m s^{-1} , but escape from the Didymos system depends on the primary’s mass and the separation, which we have not verified against a primary source. Material fast enough to leave Dimorphos may still be bound to the binary and return. A quantitative ejecta-mass likelihood is conditional on an observation operator that reconciles them. The present model β is projected along impactor momentum, whereas the orbital observation concerns Dimorphos’s orbital velocity direction (Figure 6). Both projections must be computed with verified orbital constants before physical posterior interpretation [post-processing protocol].

The predeclared relative discrepancy terms on $b = \beta - 1$ are 0.013 (realisation), 0.004 (far-field resolution), 0.053 (near-field resolution), 0.016 (late-time plateau), 0.009 (target shape),

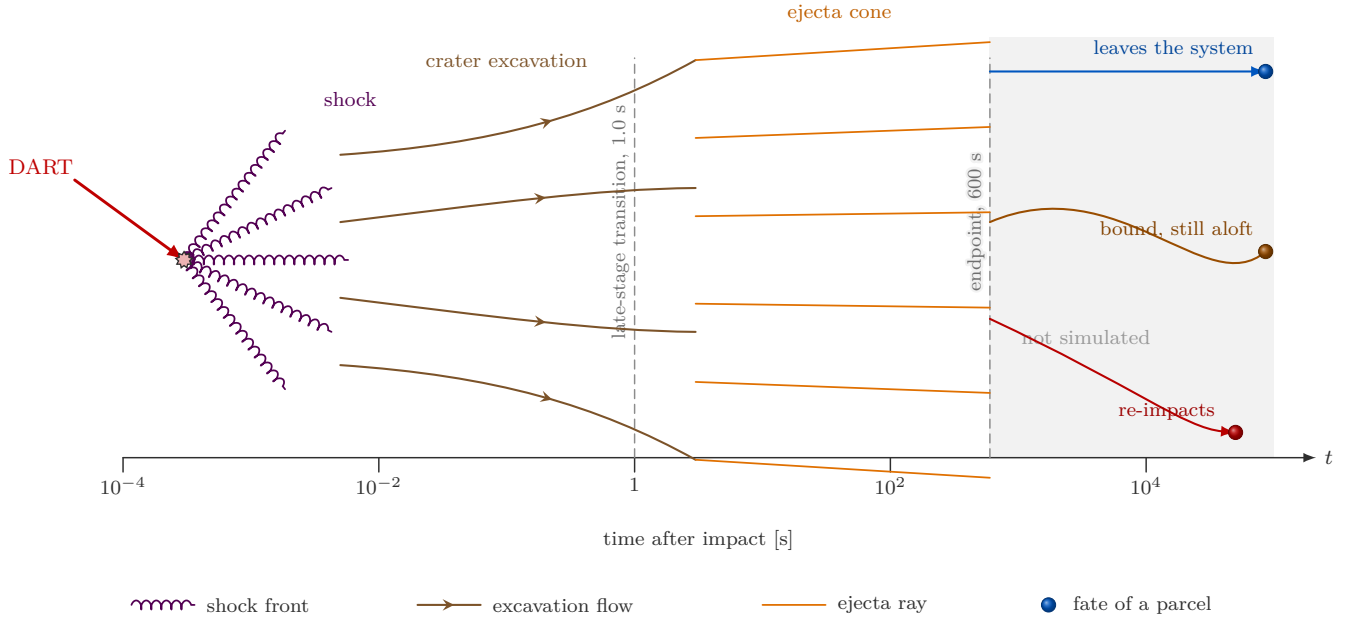


Figure 4. What a run covers, and where it stops. The shock is carried on the refined grid, the excavation flow develops through the transition time fixed at 1.0 s, and the ejecta cone is read at the 600 s endpoint. Everything to the right of that endpoint is outside the simulation: whether a parcel leaves the system, stays aloft, or returns to the surface is decided on timescales we do not integrate. The horizontal axis is logarithmic in time; vertical position is schematic.

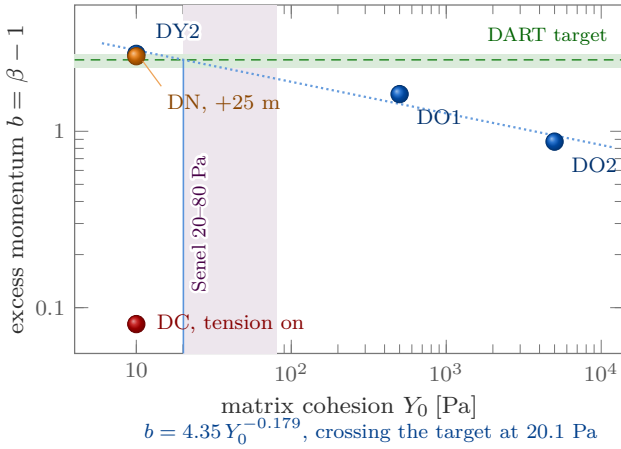


Figure 5. Where the five completed runs sit in the cohesion-momentum landscape. The dotted line is the predeclared three-point power law, refitted here from the tabulated values alone ($b = 4.35 Y_0^{-0.179}$, maximum relative residual 0.1198); the vertical line marks where it crosses the mass-adjusted DART target. That crossing is an interpolation inside one fixed scene, not a measurement of Dimorphos's strength. The shaded column is the homogeneous cohesion range reported by [5], shown for orientation only: it was obtained with a different code and a different internal friction. DC and DN share $Y_0 = 10$ Pa with DY2 and are displaced only by the change each one makes.

and 0.030 (impact location). The near-field value is documented but its executable constant is still nan; the code rejects that configuration. We therefore do not quote a complete operational denominator. The omission is not a rounding detail.

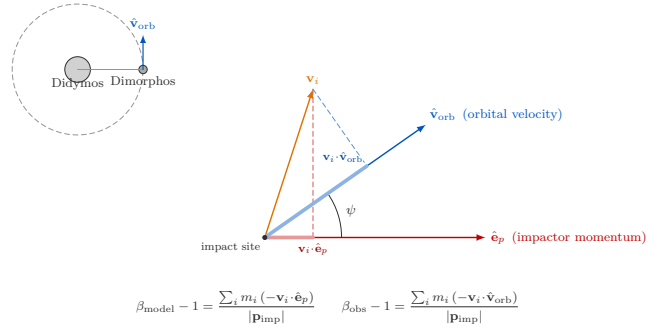


Figure 6. The simulated and the observed momentum enhancement are not the same projection. Each escaping parcel contributes its component along the impactor momentum $\hat{\mathbf{e}}_p$ to the quantity our runs report, and its component along Dimorphos's orbital velocity $\hat{\mathbf{v}}_{\text{orb}}$ to the quantity the period change constrains. We do not quote ψ : it follows from orbital constants we have not yet checked against a primary source, and the inset only fixes which direction $\hat{\mathbf{v}}_{\text{orb}}$ is, not its magnitude.

Since σ_{model}^2 is a sum of squared terms, the declared values make that one term 66.4% of the variance: σ_{model} would read 0.0377 without it against 0.0650 with it, so a denominator quoted in its absence would tighten the comparison by a factor of 1.72. The 0.134 impactor-geometry and 0.971 matrix-tension contrasts are alternative-model sensitivities, not Gaussian noise terms [discrepancy protocol].

4.3 Checks before a posterior claim

Simulation-based calibration uses at least 200 replicates with the actual surrogate and error model; a biased rank distribution blocks posterior publication [6]. Local Fisher information is evaluated at no fewer than five parameter locations. Eigenvalues above 36 count learned directions, and the smallest count over the registered correlation sensitivities governs the claim [identifiability protocol]. Only two quantitative observation groups are currently locked, so local rank cannot exceed two. The third candidate, ejecta-speed distribution, has no locked DART value and uncertainty.

A shift of more than 0.50 dex in the Y_0 mode under the declared prior alternatives prevents a single Y_0 estimate. Independent history matching excludes points with implausibility $I > 3$ [7]. The preregistered overlap and volume gates decide whether a posterior, a broader matching region, or neither is publishable. These gates are not adjusted after inspection of the production pool. If less than 90% of the 68% posterior mass lies in the compatible region, both inferential regions are withheld.

5 Discussion and conclusions

The pilot set supports three controlled findings. **First**, simulated momentum enhancement and ejecta mass decrease across the tested cohesion values in one fixed DART scene. **Second**, the selected matrix-tension toggle changes excess momentum by 97.1% and reduces the model-reported ejecta statistic by about 7.1×10^4 in that scene. **Third**, a 25 m impact offset changes β by only 2.2% while changing ejecta mass, opening angle, and median ejecta time more. These are model sensitivities, not a reconstructed interior.

Figure 7 puts every lever we have measured on one scale, and the ordering is the uncomfortable part of this paper. The single largest measured effect on b is not one of the coordinates we are trying to infer: it is the matrix-tension law, a constitutive choice we fixed by decision, at 0.971. It exceeds the entire 10–5000 Pa cohesion span, 0.682, that the inference is supposed to resolve. Impactor geometry, at 0.134, also sits above the DART observation’s own 1σ width on b . Four further choices fall below that width and are, on this evidence, not worth arguing about. A posterior over (α_b, Y_0, f) is therefore only as meaningful as the fixed choices standing above it, which is why this paper registers those choices rather than reporting a posterior.

Relation to published interior studies

Two recent studies bound what a single β can say about an interior. Raducan and colleagues ran forward simulations across assumed material sets and interior structures and compared them with the observations, favouring a low bulk density and a modest near-surface boulder fraction [4]. Senel and colleagues then reported that several distinct interiors reproduce the reference β : a homogeneous target with cohesion in the 20–80 Pa range, a double-layered interior, and a three-layered interior with boulders clustered at the impact site [5]. That is a

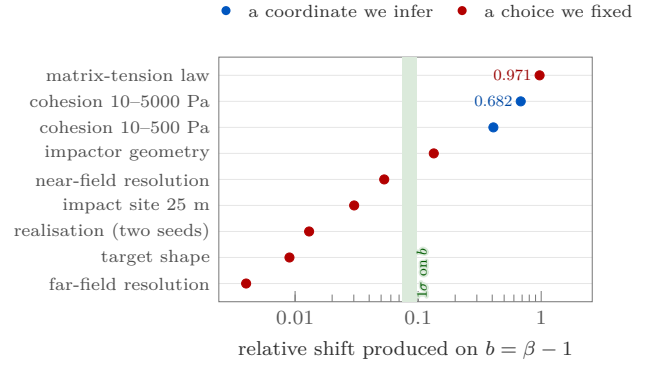


Figure 7. Every measured lever on $b = \beta - 1$, on one logarithmic scale, ranked. Blue marks a coordinate the inference is meant to learn; red marks a modelling choice we fixed. The shaded band is the observation’s own 1σ width on b , so points to its left move β by less than the measurement can resolve. The two largest entries come from single controlled pairs at one parameter location and bound the sensitivity there; they are not global derivatives.

degeneracy statement, and it is the reason the production step here is organised around how many directions in (α_b, Y_0, f) the observation actually constrains rather than around a point estimate.

Our scene-conditional interpolation, 20.1 Pa, falls at the lower edge of the 20–80 Pa homogeneous range of [5]. We do not treat that as corroboration. The two calculations differ in code, in target geometry, in dimensionality, and — importantly — in internal friction, which [5] reports as a strong lever on β : their homogeneous range is quoted at friction 0.5, whereas every run in Table 1 uses 0.6. Agreement between two numbers obtained at different friction may be coincidental. We have not measured our own sensitivity to that coefficient, and until we do, the comparison is a consistency note and nothing more.

The principal limits are structural. The three-point cohesion fit is conditional on the pilot scene. The near-field discrepancy has not been connected in code, and that omission is not isolated: the repository’s defect log now records three modules that were written, documented and unit-tested but never called from the production path, one of which had been dropped from the posterior definition before being reinstated [defect A116]. A passing test suite and a wired production path are not the same property, and we have no automated check for the second. The plume and model-ejecta definitions differ, as do the observed and simulated momentum-projection axes. Crater depth at the present 600 s endpoint is invalid or physically mismatched to a final settled crater in the pilot diagnostics [crater diagnosis]. A single earlier resolution step measured 5.34% on $\beta - 1$ in a different scene; it is not a convergence proof [resolution record].

The Hera protocol defines quantile and hash-sealing procedures, but the required posterior and validated crater/shape post-processing are not yet available [Hera protocol]. No numerical Hera forecast or production posterior is reported here.

Data and protocol provenance

The pilot values and locked decisions are archived at commit a35dab4. Figures 1 and 7 use the completed run diagnostics in Table 1 and the [locked pilot record]; Figure 2 uses the locked design coordinates. Later analyses must retain these dated decision rules. The article text and original figures are © 2026 the authors, licensed under CC BY 4.0; the Ethosoft mark, ORCID iD icon, and GitHub mark are excluded. Repository software remains under the MIT license.

Prospective Hera comparison

Hera will measure Dimorphos’s mass, map its crater and surface, and investigate its interior [8]. The agency reports arrival in November 2026, a month earlier than originally planned [9]. Arrival is not measurement, and measurement is not a published number. The characterisation phases are staggered over months, the CubeSats are released later still, and the comparison that matters here needs reduced, peer-reviewed quantities rather than a first image. DART itself is the available calibration: the impact was in September 2022, the orbital-period result was announced about two weeks later, and the momentum-transfer papers appeared roughly five months after that [2]. A sealed prediction is therefore on a different clock for each quantity, and all of those clocks run longer than the arrival date suggests. That is a scheduling constraint on this project, not a scientific result, and we do not rely on a particular date in what follows. Table 2 separates pilot evidence from future checks.

Table 2. Pilot claims and possible Hera checks. No numerical forecast is sealed yet.

Question	Evidence and independent check
Cohesion	β falls $3.7480 \rightarrow 1.8738$ and M_{ej} falls $26.1 \times$ over 10–5000 Pa. Hera’s mass measurement refines β_{obs} ; crater and shape can test later predictions. The 20.1 Pa fit is scene-conditional.
Matrix tension	The tested toggle changes β $3.7480 \rightarrow 1.0807$. Hera’s interior and surface data can constrain competing models; one toggle cannot prove zero real tensile strength.
Impact site	A 25 m offset changes β by -2.2% and M_{ej} by $+18.2\%$. Hera can map the crater and site, but cannot remeasure the past ejecta time t_{50} .

A two-paper series

Paper I (this one). Five validated pilot runs, their quantitative sensitivities, and the registered inference rules.

Paper II (planned). The production posterior or matching region, its uncertainty analysis, and numerical Hera predictions, if the claim gates of Section 4.3 pass.

The two are separate records rather than two versions of one, because Paper II reports exactly what Paper I declines to report; folding them into a single versioned record would let the later numbers overwrite the earlier refusal to state them,

which is the part worth preserving. No identifier is assigned here. A discriminating Hera comparison requires time-stamped predictions and judgment rules fixed before the relevant measurements are inspected, and the dated protocol commits cited throughout are what carry that claim until an archival record exists.

References

- [1] T. R. Daly et al., Successful kinetic impact into an asteroid for planetary defence, *Nature* **616**, 443–447 (2023). doi:10.1038/s41586-023-05810-5.
- [2] A. F. Cheng et al., Momentum transfer from the DART mission kinetic impact on asteroid Dimorphos, *Nature* **616**, 457–460 (2023). doi:10.1038/s41586-023-05878-z.
- [3] R. Lolachi et al., Ejecta Mass Estimates from the DART Impact Plume Inferred from LICIACube Images, *Planetary Science Journal* **6**, 199 (2025). doi:10.3847/PSJ/adec6b.
- [4] S. D. Raducan et al., Physical properties of asteroid Dimorphos as derived from the DART impact, *Nature Astronomy* **8**, 445–455 (2024). doi:10.1038/s41550-024-02200-3.
- [5] C. Senel et al., DART-scale Impacts into a Complex Low-strength Target: Insights into Asteroid Dimorphos’s Interior from Simulations, *Planetary Science Journal* **6**, 157 (2025). doi:10.3847/PSJ/addf31.
- [6] S. Talts et al., Validating Bayesian Inference with Simulation-Based Calibration, arXiv:1804.06788 (2018).
- [7] I. Vernon, M. Goldstein, and R. G. Bower, Galaxy Formation: Bayesian History Matching for the Observable Universe, *Statistical Science* **29**, 81–90 (2014). doi:10.1214/12-STS412.
- [8] European Space Agency, *Hera’s Science* (accessed 10 October 2026). esa.int.
- [9] European Space Agency, *ESA’s Hera targets early arrival at Didymos asteroids* (accessed 10 October 2026). esa.int.